

L1-Norm-Based Common Spatial Patterns

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Abstract—Common spatial patterns (CSP) is a commonly used method of spatial filtering for multichannel electroencephalogram (EEG) signals. The formulation of the CSP criterion is based on variance using L2-norm, which implies that CSP is sensitive to outliers. In this paper, we propose a robust version of CSP, called CSP-L1, by maximizing the ratio of filtered dispersion of one class to the other class, both of which are formulated by using L1-norm rather than L2-norm. The spatial filters of CSP-L1 are obtained by introducing an iterative algorithm, which is easy to implement and is theoretically justified. CSP-L1 is robust to outliers. Experiment results on a toy example and datasets of BCI competitions demonstrate the efficacy of the proposed method.

Index Terms—Brain-computer interfaces, common spatial patterns (CSP), CSP-L1, L1-norm, robust modeling.

I. INTRODUCTION

BRAIN-COMPUTER interfaces (BCI) have emerged as a promising way of nonmuscular communication with external world for severely paralyzed persons [29]. Electroencephalogram (EEG)-based BCI transfers intents of an individual, reflected in distinguishable EEG signals directly, into control commands for an assistive device. The successful decoding of the mental tasks heavily relies on a robust classification of the EEG signals. A large number of modern machine learning techniques have been applied in the BCI community [2], [8], [19], [23].

Common spatial patterns (CSP) is a widely used technique of spatial filtering for multivariate EEG time series in BCI [5], [7]. Given two classes of mental conditions, CSP finds a small number of spatial filters to maximize the variance of one class and meanwhile minimize the variance of the other class. CSP has been extensively investigated in BCI [12], [13], [16], [20], [24]. Computationally, CSP is solved by simultaneously diagonalizing the two covariance matrices of the two classes. Alternatively, CSP can be formulated as an optimization problem that maximizes (or minimizes) the ratio of the filtered variance of one class

to the other class, which can be solved as a generalized eigenvalue problem. Many extensions to CSP were developed in literature [5], such as common spatio-spectral patterns (CSSP) [15], local temporal common spatial patterns (LTCSP) [27], regularized CSP (RCSP) [18], and sparse CSP (SCSP) [1].

While CSP is an effective approach to discovering discriminative spatial filters, CSP is sensitive to outliers, since CSP is formulated by using the variances [5], [18]. The utilization of L2- or Frobenius-norm in the variance expression magnifies the effect of outliers that are associated with large deviations, and thus makes outliers contribute too much. Even a single contaminated sample can dramatically change the subspace spanned by the generalized eigenvectors and thus severely distort the overall solution of CSP. As a result, the spatial filters obtained are basically meaningless. Moreover, the variances are commonly used features in CSP-based classification, which likewise are prone to being affected by the outliers.

The scalp-recorded EEG signals, however, are often polluted due to outliers caused by electromyogram (EMG), electrooculogram (EOG), and spikes, as well as a large number of spontaneous brain activities. Detecting and then removing outliers in the multichannel EEG data is not a trivial issue. So, it is very beneficial to systematically develop a technique to address the robustness of the CSP algorithm with the presence of outliers, such as [3], [10], [11], [30].

In this paper, we contribute a robust version of the CSP algorithm by using L1-norm rather than L2-norm. The proposed approach, termed CSP-L1, is motivated by the basic idea of L1-norm-based modeling developed recently in the field of machine learning [14], [17], [22], where it was shown that L1-norm can effectively suppress the dominating influence of outliers. The CSP-L1 method maximizes the ratio of the L1-dispersion of one class to the other class, and vice versa. It is worthwhile to highlight three important advantages of CSP-L1 as follows: 1) CSP-L1 serves as a robust alternative to the conventional CSP. This is achieved by using the new L1-norm-based criterion. 2) A simple and efficient iterative algorithm is introduced to find the spatial filters, which is theoretically shown to arrive at a locally maximal solution. 3) The L1-dispersion is used as features for classification.

The remainder of this paper is organized as follows. In Section II, the conventional CSP method is briefly reviewed. In Section III, we propose the CSP-L1 method, for which an iterative algorithm is presented and theoretically justified. The experimental results are summarized in Section IV. Finally, Section V concludes the paper.

II. BRIEF REVIEW OF CONVENTIONAL CSP

In the context of EEG signal processing, the CSP method is applied to two-class paradigm. Let $\mathbf{X}^1, \dots, \mathbf{X}^{t_x} \in \mathbb{R}^{c \times l}$ be

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the trial segments of EEG signals that belong to one condition, and $\mathbf{Y}^1, \dots, \mathbf{Y}^{t_y} \in \mathbb{R}^{c \times l}$ be the trial segments of the other condition, where c denotes the number of channels (i.e., recording electrodes), l is the number of samples (i.e., recording time points) in a trial, and t_x and t_y are the numbers of trials of the two classes respectively. The trial segments are assumed to be already band-pass filtered, centered and scaled. Specifically, each raw trial, denoted using a general symbol $\mathbf{Z}_{\text{raw}}$, from either class is first band-pass filtered resulting in $\mathbf{Z}_{\text{band-pass}}$, then center $\mathbf{Z}_{\text{band-pass}}$ by subtracting per-channel mean resulting in $\mathbf{Z}_{\text{centered}}$, and finally scaled as $\mathbf{Z} = \mathbf{Z}_{\text{centered}} / \sqrt{\text{tr}(\mathbf{Z}_{\text{centered}} \mathbf{Z}_{\text{centered}}^T)}$, where tr is the trace operator of matrix.

Write $\mathbf{X} = (\mathbf{X}^1, \dots, \mathbf{X}^{t_x})$ and $\mathbf{Y} = (\mathbf{Y}^1, \dots, \mathbf{Y}^{t_y})$. For simplicity of expression, relabel the columns of $\mathbf{X}$ and $\mathbf{Y}$ as $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_m) \in \mathbb{R}^{c \times m}$ and $\mathbf{Y} = (\mathbf{y}_1, \dots, \mathbf{y}_n) \in \mathbb{R}^{c \times n}$, respectively. Here, m (equal to $l \times t_x$) and n (equal to $l \times t_y$) are the total numbers of sampled points of the two classes. The observation at a specific time point is a vector in the c -dimensional Euclidean space. The spatial covariances of the two classes are calculated as $\mathbf{C}_x = \frac{1}{t_x} \mathbf{X} \mathbf{X}^T$ and $\mathbf{C}_y = \frac{1}{t_y} \mathbf{Y} \mathbf{Y}^T$.

The CSP approach aims to find a spatial filter $\omega \in \mathbb{R}^c$ such that the variance of the projected data points of one class is maximized while the other class is minimized. Mathematically, the spatial filter of CSP is formulated by maximizing (or minimizing) the criterion [5], [23]

$$J(\omega) = \frac{\omega^T \mathbf{C}_x \omega}{\omega^T \mathbf{C}_y \omega}. \quad (1)$$

The spatial filter is solved by the generalized eigenvalue equation

$$\mathbf{C}_x \omega = \lambda \mathbf{C}_y \omega. \quad (2)$$

The eigenvalue λ measures the ratio of variances of the two classes. For the purpose of classification, we select a small number of filters that are the generalized eigenvectors associated with eigenvalues from both ends of the eigenvalue spectrum. The variances (possibly after a log-transformation) of the spatially filtered EEG data are used as features for classification.

III. L1-NORM-BASED CSP

The CSP approach models EEG data using variances that are based on L2-norm. To see this, we rewrite (1) as

$$J(\omega) = \frac{\omega^T \mathbf{C}_x \omega}{\omega^T \mathbf{C}_y \omega} = \frac{\frac{1}{t_x} \|\omega^T \mathbf{X}\|_2^2}{\frac{1}{t_y} \|\omega^T \mathbf{Y}\|_2^2} = \frac{\frac{1}{t_x} \sum_{i=1}^m (\omega^T \mathbf{x}_i)^2}{\frac{1}{t_y} \sum_{j=1}^n (\omega^T \mathbf{y}_j)^2} \quad (3)$$

where $\|\cdot\|_2$ denotes L2-norm. It is well known that L2-norm is sensitive to outliers, since the square operator will magnify the effect of large deviations. On the other hand, EEG signals are usually contaminated by noise or artifact. These atypical observations may badly affect the desired solution of CSP. From the perspective of statistical modeling, L1-norm is more robust than L2-norm. Motivated by the basic idea of L1-norm-based modeling, we propose to maximize (or minimize) the CSP-like

criterion using L1-norm instead of L2-norm as follows:

$$\tilde{J}(\omega) = \frac{\|\omega^T \mathbf{X}\|_1}{\|\omega^T \mathbf{Y}\|_1} = \frac{\sum_{i=1}^m |\omega^T \mathbf{x}_i|}{\sum_{j=1}^n |\omega^T \mathbf{y}_j|} \quad (4)$$

where $\|\cdot\|_1$ denotes L1-norm. We refer to (4) as CSP-L1. The absolute value operation is not differentiable, which makes the optimization of the objective functions difficult. In the following, we present an iterative algorithm to solve the spatial filter ω that maximizes the objective function (4). The minimization of (4) can be likewise solved by maximizing $1/\tilde{J}(\omega)$.

A. Algorithmic Procedure of CSP-L1

We use t to denote the iteration number, and assume that $\omega \neq 0$.

- 1) Set $t = 0$. Initialize $\omega(t)$ by any a c -dimensional vector followed, and rescale it to unit length.
- 2) Define two polarity functions $p_i(t)$ and $q_j(t)$ taking either 1 or -1 to implement the absolute value operation in (4):

$$p_i(t) = \text{sgn}(\omega^T(t) \mathbf{x}_i) \quad (5)$$

and

$$q_j(t) = \text{sgn}(\omega^T(t) \mathbf{y}_j) \quad (6)$$

where $\text{sgn}(\cdot)$ is the sign function.

- 3) Let

$$\mathbf{d}(t) = \frac{\sum_{i=1}^m p_i(t) \mathbf{x}_i}{\sum_{i=1}^m |\omega^T(t) \mathbf{x}_i|} - \frac{\sum_{j=1}^n q_j(t) \mathbf{y}_j}{\sum_{j=1}^n |\omega^T(t) \mathbf{y}_j|}. \quad (7)$$

Update $\omega(t)$ by

$$\omega(t+1) = \omega(t) + \eta \mathbf{d}(t) \quad (8)$$

where η is a parameter of learning rate taking a small positive value, and rescale $\omega(t+1)$ to unit length. In our experiments on real EEG data, we try a set of different values (including zero) of η , and select the update vector resulting in the largest value of the objective function. With this constraint, we avoid the case that $\omega(t+1)$ overshoots the optimal point that leads to a smaller value of the objective function. Set $t \leftarrow t + 1$.

- 4) If $\tilde{J}(\omega(t))$ can not increase significantly (measured by increase speed), then stop the iteration and set $\omega^* = \omega(t)$. Otherwise, go to Step 2.
- 5) Output ω^* as the spatial filter.

The computational complexity of each iteration of the above algorithm is $O((m+n)c)$. The practical computational cost involves the number of iterations at convergence, which however depends on the EEG data at hand. In our experiments on real EEG data, we observe that the optimization converges in 50 steps.

B. Justification

In this section, we show the following justification.

Theorem: With the above CSP-L1 iterative procedure, the objective function $\tilde{J}(\omega(t))$ is a nondecreasing function at each step of iteration t .

Proof. We commence with considering (4). At iteration t , we have that

$$\tilde{J}(\omega(t)) = \frac{\sum_{i=1}^m |\omega^T(t) \mathbf{x}_i|}{\sum_{j=1}^n |\omega^T(t) \mathbf{y}_j|}. \quad (9)$$

By using the definition of $p_i(t)$, the numerator of (9) can be rewritten as

$$\sum_{i=1}^m |\omega^T(t) \mathbf{x}_i| = \omega^T(t) \left(\sum_{i=1}^m p_i(t) \mathbf{x}_i \right) = \omega^T(t) \mathbf{u}(t) \quad (10)$$

where

$$\mathbf{u}(t) = \sum_{i=1}^m p_i(t) \mathbf{x}_i. \quad (11)$$

The denominator of (9) can be rewritten as

$$\begin{aligned} \sum_{j=1}^n |\omega^T(t) \mathbf{y}_j| &= \frac{1}{2} \omega^T(t) \left(\sum_{j=1}^n \frac{\mathbf{y}_j \mathbf{y}_j^T}{|\omega^T(t) \mathbf{y}_j|} \right) \omega(t) \\ &+ \frac{1}{2} \sum_{j=1}^n |\omega^T(t) \mathbf{y}_j| = \frac{1}{2} \omega^T(t) \mathbf{V}(t) \omega(t) + \frac{1}{2} \|\mathbf{z}(t)\|_1 \end{aligned} \quad (12)$$

where

$$\mathbf{V}(t) = \sum_{j=1}^n \frac{\mathbf{y}_j \mathbf{y}_j^T}{|z_j(t)|} \quad (13)$$

$z_j(t) = \omega^T(t) \mathbf{y}_j$, and $\mathbf{z}(t)$ is the vector having the entries $\{z_j(t)\}_{j=1,\dots,n}$. Here, $\|\mathbf{z}(t)\|_1$ is the L1-norm of $\mathbf{z}(t)$, i.e., $\|\mathbf{z}(t)\|_1 = \sum_{j=1}^n |z_j(t)|$. Substituting (10) and (12) into (9), we have that

$$\tilde{J}(\omega(t)) = \frac{\omega^T(t) \mathbf{u}(t)}{\frac{1}{2} \omega^T(t) \mathbf{V}(t) \omega(t) + \frac{1}{2} \|\mathbf{z}(t)\|_1}. \quad (14)$$

Since direct derivative of $\tilde{J}$ is intractable, as an intermediate step, we introduce a surrogate function

$$L(\xi) = \frac{\xi^T \mathbf{u}(t)}{\frac{1}{2} \xi^T \mathbf{V}(t) \xi + \frac{1}{2} \|\mathbf{z}(t)\|_1}. \quad (15)$$

Note that $L(\xi)$ is a function of variable ξ . We keep in mind that $\omega(t)$ is a constant vector at step t . So, in the function $L(\xi)$, only ξ is the variable while other quantities are fixed values. We should not confuse the variable ξ with the constant $\omega(t)$. We calculate the gradient of $L(\xi)$ with respect to ξ as

$$\begin{aligned} \mathbf{g}(\xi) &= \frac{\partial L(\xi)}{\partial \xi} \\ &= \frac{(\frac{1}{2} \xi^T \mathbf{V}(t) \xi + \frac{1}{2} \|\mathbf{z}(t)\|_1) \mathbf{u}(t) - (\xi^T \mathbf{u}(t)) \mathbf{V}(t) \xi}{(\frac{1}{2} \xi^T \mathbf{V}(t) \xi + \frac{1}{2} \|\mathbf{z}(t)\|_1)^2}. \end{aligned} \quad (16)$$

The gradient value at the point $\omega(t)$ is obtained by substituting $\omega(t)$ into the function $\mathbf{g}(\xi)$:

$$\mathbf{g}(\omega(t)) = \frac{\sum_{i=1}^m p_i(t) \mathbf{x}_i}{\sum_{j=1}^n |\omega^T(t) \mathbf{y}_j|}$$

$$\begin{aligned} &= \frac{(\sum_{i=1}^m |\omega^T(t) \mathbf{x}_i|)(\sum_{j=1}^n q_j(t) \mathbf{y}_j)}{(\sum_{j=1}^n |\omega^T(t) \mathbf{y}_j|)^2} \\ &\propto \frac{\sum_{i=1}^m p_i(t) \mathbf{x}_i}{\sum_{i=1}^m |\omega^T(t) \mathbf{x}_i|} - \frac{\sum_{j=1}^n q_j(t) \mathbf{y}_j}{\sum_{j=1}^n |\omega^T(t) \mathbf{y}_j|} \end{aligned} \quad (17)$$

which is the exact $\mathbf{d}(t)$ given in (7). That is, $\mathbf{d}(t)$ is the vector that points to the ascending direction of $L(\xi)$ at the point $\omega(t)$. Again, note that $L(\xi)$ is a function of variable ξ . Thus, with the vector $\omega(t+1)$ given by (8), and replacing the variable ξ with $\omega(t)$ and $\omega(t+1)$, we have that $L(\omega(t)) \leq L(\omega(t+1))$, i.e.,

$$\begin{aligned} &\frac{\omega^T(t) \mathbf{u}(t)}{\frac{1}{2} \omega^T(t) \mathbf{V}(t) \omega(t) + \frac{1}{2} \|\mathbf{z}(t)\|_1} \\ &\leq \frac{\omega^T(t+1) \mathbf{u}(t)}{\frac{1}{2} \omega^T(t+1) \mathbf{V}(t) \omega(t+1) + \frac{1}{2} \|\mathbf{z}(t)\|_1}. \end{aligned} \quad (18)$$

Note that both the left side of (18) and the denominator of the right side are non-negative, which implies that the numerator of the right side is also non-negative. It is known that directly differentiating the objective function $\tilde{J}(\omega)$ is intractable. To circumvent this obstacle, an elegant trick used here is to introduce the differentiable function $L(\xi)$ and obtain the transitional inequality (18). The purpose of (18) is to relate $\tilde{J}(\omega(t))$ and $\tilde{J}(\omega(t+1))$. As will be seen, by the bridging of (18), the final monotony property of our algorithm is established.

We consider the numerator of (18). Since $p_i(t+1)$ is the polarity of $\omega^T(t+1) \mathbf{x}_i$, the quantity $p_i(t+1) \omega^T(t+1) \mathbf{x}_i$ is always non-negative for $i = 1, \dots, m$. However, for $p_i(t)$, it may happen that $p_i(t) \omega^T(t+1) \mathbf{x}_i$ is negative for some i . Therefore, we have that

$$\begin{aligned} 0 &\leq \omega^T(t+1) \mathbf{u}(t) = \omega^T(t+1) \sum_{i=1}^m p_i(t) \mathbf{x}_i \\ &\leq \omega^T(t+1) \sum_{i=1}^m p_i(t+1) \mathbf{x}_i \\ &= \sum_{i=1}^m |\omega^T(t+1) \mathbf{x}_i|. \end{aligned} \quad (19)$$

We proceed to consider the denominator of (18). We have that

$$\begin{aligned} &\frac{1}{2} \omega^T(t+1) \mathbf{V}(t) \omega(t+1) + \frac{1}{2} \|\mathbf{z}(t)\|_1 \\ &= \frac{1}{2} \sum_{j=1}^n \frac{(\omega^T(t+1) \mathbf{y}_j)^2}{|z_j(t)|} + \frac{1}{2} \|\mathbf{z}(t)\|_1 \\ &\geq \min_{\zeta \in \mathbb{R}_+^n} \frac{1}{2} \sum_{j=1}^n \frac{(\omega^T(t+1) \mathbf{y}_j)^2}{\zeta_j} + \frac{1}{2} \|\zeta\|_1 \\ &= \sum_{j=1}^n |\omega^T(t+1) \mathbf{y}_j| \end{aligned} \quad (20)$$

where $\{\zeta_j\}_{j=1,\dots,n}$ are the entries of ζ , and the last equation holds is due to the fact as follows.

Lemma: For any vector $\mathbf{a} = (a_1, \dots, a_n)^T \in \mathbb{R}^n$, the following variational equality holds [9]

$$\|\mathbf{a}\|_1 = \min_{\zeta \in \mathbb{R}_+^n} \frac{1}{2} \sum_{j=1}^n \frac{a_j^2}{\zeta_j} + \frac{1}{2} \|\zeta\|_1 \quad (21)$$

and the minimum is uniquely reached at $\zeta_j = |a_j|$ for $j = 1, \dots, n$, where $\zeta = (\zeta_1, \dots, \zeta_n)^T$.

Substituting (19) and (20) into (18), we have that

$$\begin{aligned} & \frac{\omega^T(t+1)\mathbf{u}(t)}{\frac{1}{2}\omega^T(t+1)\mathbf{V}(t)\omega(t+1) + \frac{1}{2}\|\mathbf{z}(t)\|_1} \\ & \leq \frac{\sum_{i=1}^m |\omega^T(t+1)\mathbf{x}_i|}{\sum_{j=1}^n |\omega^T(t+1)\mathbf{y}_j|} \\ & = \tilde{J}(\omega(t+1)). \end{aligned} \quad (22)$$

Combining (14), (18), and (22), we have that

$$\tilde{J}(\omega(t)) \leq \tilde{J}(\omega(t+1)). \quad (23)$$

Since $\tilde{J}(\omega(t+1))$ is invariant with the scale of $\omega(t+1)$, i.e., $\tilde{J}(\omega(t+1)) = \tilde{J}(k\omega(t+1))$, where k is any a scale constant. Thus, we can normalize $\omega(t+1)$ as $\omega(t+1)/\|\omega(t+1)\|_2$. That is, only the direction of the projection vector is of interest. The proof is thus completed. ■

Theoretically, we prove that $\tilde{J}(\omega(t))$ is nondecreasing with respect to t . In practice, we usually observe that $\tilde{J}(\omega(t))$ is monotonically increasing, since the trivial case that $\tilde{J}$ remains the same value with updated $\omega(t)$ seldom happens. The theorem guarantees that the vector $\omega(t)$ moves toward a local maximum solution of (4). The global maximum solution, however, is not necessarily to be reached. Considering that the initial vector $\omega(0)$ can be specified arbitrarily, we can set it appropriately, for example, as the basis vectors of the conventional CSP. With this setting, CSP-L1 can be viewed as modifying the basis vectors of CSP toward the L1-dispersion optimization by each iteration. In another way, we can try different initial vectors to run the CSP-L1 algorithm, and output the projection vector that results in the maximum $\tilde{J}$ value.

C. Extension to Multiple Spatial Filters

Once the first spatial filter ω_1 that maximizes the objective function (4) is obtained, we may proceed to find the second spatial filter ω_2 . We require that ω_2 maximizes (4) subject to the constraint $\omega_1^T \omega_2 = 0$. In other words, we seek ω_2 in the orthogonally complementary space of ω_1 . Therefore, ω_2 can be represented as

$$\omega_2 = (\mathbf{I}_c - \omega_1 \omega_1^T) \alpha_1 \quad (24)$$

where α_1 is a c -dimensional coefficient vector. Substituting (24) into (4), we find α_1 to maximize

$$\frac{\sum_{i=1}^m |\alpha_1^T (\mathbf{I}_c - \omega_1 \omega_1^T) \mathbf{x}_i|}{\sum_{j=1}^n |\alpha_1^T (\mathbf{I}_c - \omega_1 \omega_1^T) \mathbf{y}_j|}. \quad (25)$$

Let

$$\mathbf{x}_i^{(2)} = (\mathbf{I}_c - \omega_1 \omega_1^T) \mathbf{x}_i = \mathbf{x}_i - (\omega_1^T \mathbf{x}_i) \omega_1 \quad (26)$$

$$\mathbf{y}_j^{(2)} = (\mathbf{I}_c - \omega_1 \omega_1^T) \mathbf{y}_j = \mathbf{y}_j - (\omega_1^T \mathbf{y}_j) \omega_1 \quad (27)$$

for $i = 1, \dots, m; j = 1, \dots, n$. The $\mathbf{x}_i^{(2)}$ and $\mathbf{y}_j^{(2)}$ can be interpreted as new samples, in which the information contained in the first spatial filter ω_1 is discarded. By (26) and (27), (25) can be rewritten as

$$\frac{\sum_{i=1}^m |\alpha_1^T \mathbf{x}_i^{(2)}|}{\sum_{j=1}^n |\alpha_1^T \mathbf{y}_j^{(2)}|} \quad (28)$$

which has the same form with (4) using the new samples $\mathbf{x}_i^{(2)}$ and $\mathbf{y}_j^{(2)}$. As a result, α_1 can be solved by the iterative algorithm presented in Section III-A, where $\mathbf{x}_i$ and $\mathbf{y}_j$ are replaced by $\mathbf{x}_i^{(2)}$ and $\mathbf{y}_j^{(2)}$, respectively. Then ω_2 is obtained by (24). Also, ω_2 is normalized as $\omega_2 / \|\omega_2\|_2$. Noting (24), it follows that

$$\omega_1^T \omega_2 = 0. \quad (29)$$

Generally, suppose the first k spatial filters $\omega_1, \dots, \omega_k$ have been obtained. The $(k+1)$ th spatial filter ω_{k+1} is selected to maximize (4) subject to the constraint $\omega_1^T \omega_{k+1} = \omega_2^T \omega_{k+1} = \dots = \omega_k^T \omega_{k+1} = 0$. Let $\mathbf{W}_k = (\omega_1, \dots, \omega_k)$. Then there exists one vector $\alpha_k \in \mathbb{R}^c$ such that

$$\omega_{k+1} = (\mathbf{I}_c - \mathbf{W}_k \mathbf{W}_k^T) \alpha_k \quad (30)$$

since the orthogonally complementary space of the orthonormal vectors $\omega_1, \dots, \omega_k$ is spanned by $\mathbf{I}_c - \mathbf{W}_k \mathbf{W}_k^T$. Likewise, α_k can be solved by the iterative algorithm using the updated samples

$$\mathbf{x}_i^{(k+1)} = (\mathbf{I}_c - \mathbf{W}_k \mathbf{W}_k^T) \mathbf{x}_i = (\mathbf{I}_c - \omega_k \omega_k^T) \mathbf{x}_i^{(k)} \quad (31)$$

$$\mathbf{y}_j^{(k+1)} = (\mathbf{I}_c - \mathbf{W}_k \mathbf{W}_k^T) \mathbf{y}_j = (\mathbf{I}_c - \omega_k \omega_k^T) \mathbf{y}_j^{(k)} \quad (32)$$

in which we use the equation

$$\mathbf{I}_c - \mathbf{W}_k \mathbf{W}_k^T = (\mathbf{I}_c - \omega_k \omega_k^T) (\mathbf{I}_c - \mathbf{W}_{k-1} \mathbf{W}_{k-1}^T). \quad (33)$$

So, it is feasible to obtain $\mathbf{x}_i^{(k+1)}$ and $\mathbf{y}_j^{(k+1)}$ by updating $\mathbf{x}_i^{(k)}$ and $\mathbf{y}_j^{(k)}$, i.e., multiplied by $\mathbf{I}_c - \omega_k \omega_k^T$ from the left side, respectively. The obtained spatial filter ω_{k+1} is normalized as $\omega_{k+1} / \|\omega_{k+1}\|_2$, and the transformation matrix $\mathbf{W}_k$ is padded as $\mathbf{W}_{k+1} = (\mathbf{W}_k, \omega_{k+1})$.

D. Feature Extraction

By maximizing the objective function (4), we obtain a set of orthonormal spatial filters $\omega_1, \dots, \omega_{n_x}$. Likewise, By maximizing the objective function $1/\tilde{J}(\omega)$, we obtain another set of orthonormal spatial filters $\omega'_1, \dots, \omega'_{n_y}$. For any EEG single trial $\mathbf{Z}$, the feature is extracted as

$$\mathbf{f} = \begin{pmatrix} \|\omega_1^T \mathbf{Z}\|_1 \\ \vdots \\ \|\omega_{n_x}^T \mathbf{Z}\|_1 \\ \|(\omega'_1)^T \mathbf{Z}\|_1 \\ \vdots \\ \|(\omega'_{n_y})^T \mathbf{Z}\|_1 \end{pmatrix} \quad (34)$$

where $\mathbf{f}$ is an $(n_x + n_y)$ -dimensional feature vector.

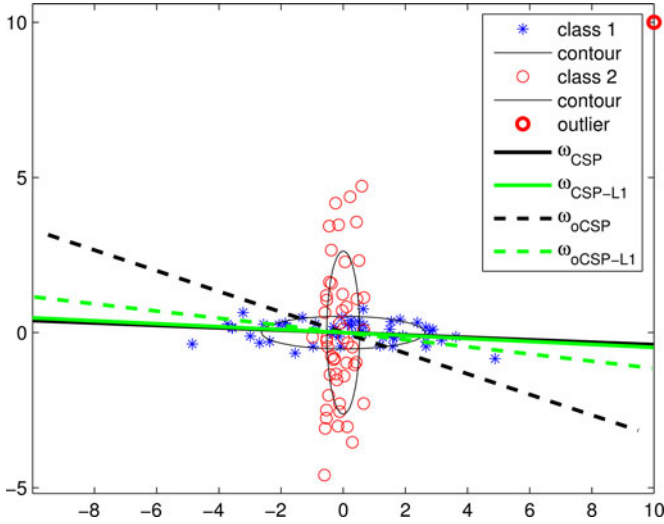


Fig. 1. A toy example of spatial filtering by using CSP-L1 and CSP on a 2-D dataset. Data points and projection vectors obtained by CSP-L1 and CSP in the cases without and with outlier were shown.

IV. EXPERIMENTS

In this section, we first used a toy dataset, where an outlier was introduced, to compare the robustness of the proposed CSP-L1 approach and the standard CSP method. Then we applied CSP-L1, the standard CSP and various CSP variants to the problem of single-trial EEG classification on two publicly available real EEG datasets: dataset IVa of BCI competition III and dataset Ila of BCI competition IV. We compared their performances when multivariate outliers occurred.

A. Toy Example

A 2-D artificial dataset was used to demonstrate the robustness of CSP-L1. The simulated data were generated from two Gaussian classes with zero means and covariance matrices being $[5, 0; 0, .2]$ and $[.2, 0; 0, 5]$. For each class, 50 data points were randomly yielded, as depicted in Fig. 1, where the two classes of samples were specified by “*” and “o.” We introduced an additional outlier, i.e. $[10, 10]$, into class “o” to test the performance of CSP-L1 and CSP in withstanding the impact of the outlier. For illustration, we applied CSP-L1 and CSP to extract the spatial filter that maximized the dispersion of class “*” and minimized the dispersion of class “o.”

If the outlier was discarded, CSP-L1 and CSP produced almost identical projection vectors, namely $\omega_{\text{CSP}} = [-0.9993, 0.0387]$ and $\omega_{\text{CSP-L1}} = [-0.9989, 0.0476]$, as shown in Fig. 1. The projection ω_{CSP} gave the largest variance in class “*” and the smallest variance in class “o.” Likewise, the projection $\omega_{\text{CSP-L1}}$ gave the largest dispersion in terms of L1-norm in class “*” and the smallest dispersion in class “o.”

If the outlier occurred, the projection vectors of CSP-L1 and CSP were $\omega_{\text{oCSP}} = [-0.9490, 0.3153]$ and $\omega_{\text{oCSP-L1}} = [-0.9934, 0.1146]$, respectively, as shown in Fig. 1. In using CSP-L1, the initial projection vector was set as the solution of CSP, and the learning rate parameter η was set as 0.01. By the

results, it was seen that CSP was badly influenced by the outlier, while CSP-L1 successfully suppressed the effect of the outlier.

The value of the CSP-L1 objective function $\tilde{J}(\omega(t))$ was enhanced by each step of iteration. In the case of outlier-free only 3 iterations made the algorithm of CSP-L1 converge while 14 iterations were needed in the case that outlier happened.

If the initial projection vector was set as the sample with the largest length, i.e., $\omega(0) = \arg \max_i \|x_i\|_2$, then CSP-L1 produced almost identical basis vectors with above results of CSP-L1. The numbers of iterations in the cases without and with outlier were 9 and 12, respectively. We tried other random initializations, and the same basis vectors were reached. So, CSP-L1 was not very sensitive with its initialization. The iteration number, however, varied with different initial values.

B. Real EEG Datasets for Evaluation

1) *Dataset IVa of BCI Competition III*: This dataset contained EEG signals recorded from five subjects using 118 electrodes [4]. In each trial, a visual cue was shown for 3.5s, during which three motor imageries were performed, i.e., left hand, right hand and right foot. The motor imageries of right hand and foot were needed to be classified. The total number of EEG trials for each subject was 280. Specifically, 168, 224, 84, 56, and 28 trials were used as training data corresponding to the five subjects: *aa*, *al*, *av*, *aw*, and *ay*, respectively. The data were band-pass filtered between 0.05 and 200 Hz and down-sampled at 100 Hz for succedent analysis.

2) *Dataset Ila of BCI Competition IV*: The EEG data were obtained by recording nine persons, who were asked to perform four different motor imagery tasks, i.e., left hand, right hand, foot, and tongue [21]. Two sessions recorded on different days were made available for each individual, and each session consisted of 288 trials with 72 trials per class. The EEG signals, measured by 22 electrodes, were sampled at 250 Hz. In our experiment, the session containing the class labels for all trials was used as training set and the unlabelled session as testing set. We focused on two-class classification scenario, i.e., classifying EEG signals belonging to left and right hand motor imageries.

C. Preprocessing

The EEG signals were band-pass filtered with cutoff frequencies 8 Hz and 35 Hz by using a fifth order Butterworth filter. Following the winner of BCI competition IV and [18], we used the EEG segments recorded from 0.5s to 3.75s after the visual cue that indicates the start of imaginary as samples on the first dataset, and used the time interval from 0.5s to 2.5s on the second dataset.

D. Introducing Outliers

We introduced multivariate outliers into the training datasets, and then compared the classification accuracies of the proposed CSP-L1 approach and various existing CSP-based methods. For each subject, the outliers were generated by the c -dimensional Gaussian distribution $N_c(\mu + 3\sigma, \Sigma)$, where μ and σ were the mean and standard deviation vectors of all channels of the

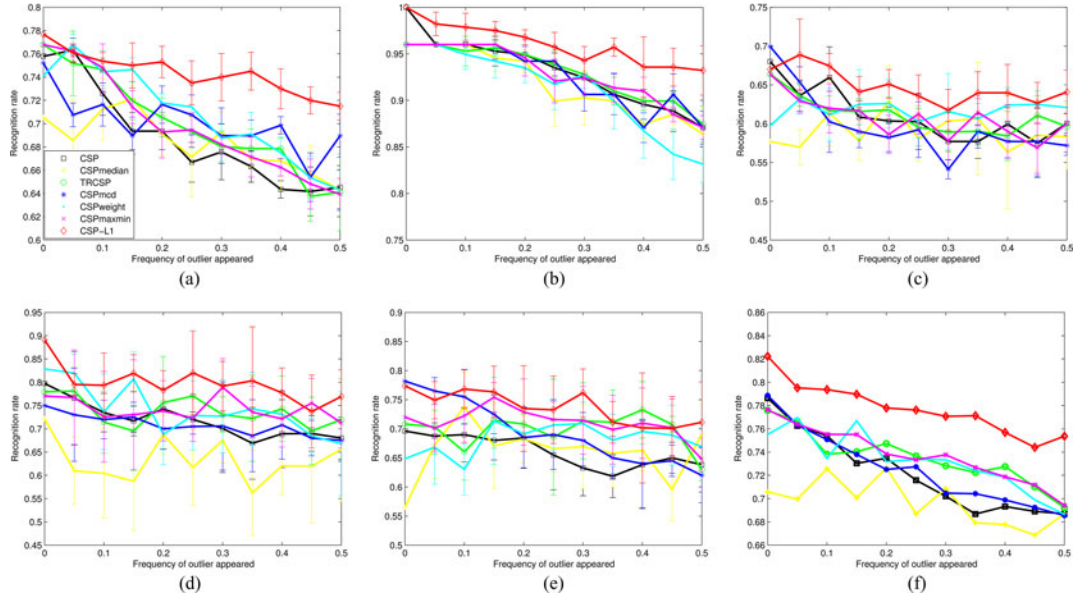


Fig. 2. Classification accuracies of the five subjects (*aa*, *al*, *av*, *aw*, and *ay* corresponding to subplots (a)–(e)) on dataset IVa of BCI competition III using various CSP variants and the proposed CSP-L1 approach with increasing occurrence frequencies of outliers. Subplot (f) showed the performance averaged across the five participants.

training EEG samples, and Σ was the covariance matrices of the training EEG samples. The generated outliers were added to the training EEG data to simulate polluted EEG signals. The number of outliers varied from 0 to $0.5(m + n)$ with step $0.05(m + n)$. We randomly selected the time positions to add the outliers.

E. Experiment Settings and Results

We compared the robustness of CSP-L1 with the standard CSP and its various variants, i.e., selecting the spatial filters of CSP by median scores (CSPmedian for short) [5], regularized CSP with Tikhonov regularization (TRCSP) [18], robust CSP with minimum covariance determinant estimator (CSPmcd) [25], [30], modified CSP with weighted average covariance matrices (CSPweight) [10], and robust CSP with maxmin approach (CSPmaxmin) [11].

The regularization parameter in TRCSP and the parameter determining the percentage of samples used to estimate the covariance matrix with minimum determinant in CSPmcd were selected by cross-validation on the training set. The weighting function used in CSPweight was the square root function. The radii of the two balls involved in CSPmaxmin were set to be equal and were selected by cross-validation, and the shape of the balls was assumed to be isotropic, i.e., the Frobenius norm was used as measure of tolerance. The initial solutions of CSP-L1 were set as the corresponding solutions of the standard CSP. In each iteration, the update (8) was run for different values of η between $1e-5$ and $9e-1$, and the updated filter with the highest value for the objective function on the training data was adopted. In other words, the learning rate was not necessarily required to be constant in iterations.

As suggested in [5] and [18], we used three pairs of spatial filters for feature extraction in using all aforementioned CSP-based methods and CSP-L1, i.e., $n_x = n_y = 3$.

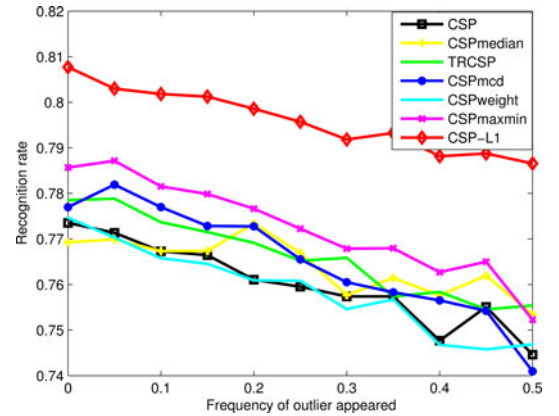


Fig. 3. Average classification accuracies across the nine subjects (*A01E* through *A09E*) on dataset IIa of BCI competition IV using various CSP variants and the proposed CSP-L1 approach with increasing occurrence frequencies of outliers.

The Fisher discriminant analysis (FDA) [28] classifier was employed to predict the condition labels of the testing EEG trials. Specifically, the resulting 6-D feature vectors after applying CSP-based methods and CSP-L1 were further projected into 1-D values by using the basis vector of FDA learned from the training features. Then, the classification of the testing EEG trials was carried out by comparing with a threshold. For each occurrence frequency of outliers, experiments were repeated 10 times, the average accuracy and the corresponding standard deviation were recorded.

The results of classification accuracies, as well as the corresponding standard deviations, on the first EEG datasets was presented in Fig. 2. The results on the second EEG dataset were presented in Fig. 3 and Table I. Some mappings of spatial filters learned by the seven CSP-related methods for subject *aa* were illustrated in Fig. 4.

TABLE I
CLASSIFICATION ACCURACIES (%) OF THE NINE SUBJECTS (*A01E* THROUGH *A09E*) ON DATASET IIA OF BCI COMPETITION IV USING VARIOUS CSP VARIANTS AND THE PROPOSED CSP-L1 APPROACH WITH INCREASING OCCURRENCE FREQUENCIES OF OUTLIERS

	Subject <i>A01E</i>							Subject <i>A02E</i>							Subject <i>A03E</i>						
Frequency	0	0.1	0.2	0.3	0.4	0.5	Mean	0	0.1	0.2	0.3	0.4	0.5	Mean	0	0.1	0.2	0.3	0.4	0.5	Mean
CSP	90.1	87.0	85.6	82.6	81.8	80.9	84.67	53.3	49.7	48.9	49.5	48.1	48.2	49.62	91.5	91.0	91.0	90.8	90.4	90.1	90.80
CSPmedian	90.6	86.6	84.8	79.3	80.7	78.2	83.37	55.9	54.0	57.6	54.1	54.1	53.6	54.88	92.0	91.7	91.1	91.2	90.9	91.2	91.35
TRCSP	90.6	88.2	86.6	83.9	80.8	83.3	85.57	53.1	50.7	49.9	49.5	49.3	48.8	50.22	92.0	92.0	91.7	91.3	90.9	90.4	91.38
CSPmcd	85.0	83.3	82.5	80.9	78.8	74.7	80.87	58.6	55.4	53.6	51.1	50.0	49.7	53.07	95.4	93.2	92.5	91.4	91.2	91.3	92.50
CSPweight	90.2	88.7	87.1	84.9	82.8	84.4	86.35	55.5	51.7	51.8	49.5	50.2	49.7	51.40	91.6	91.7	91.3	91.1	91.2	91.0	91.32
CSPmaxmin	88.8	86.1	84.8	82.6	81.4	78.8	83.75	56.8	53.7	52.2	51.3	49.9	49.6	52.25	94.5	93.0	92.8	91.8	91.8	91.7	92.60
CSP-L1	91.3	89.9	86.9	85.7	84.5	83.4	86.95	58.6	54.9	54.4	52.9	51.7	53.7	54.37	96.8	96.4	96.3	96.2	96.2	95.9	96.30
	Subject <i>A04E</i>							Subject <i>A05E</i>							Subject <i>A06E</i>						
Frequency	0	0.1	0.2	0.3	0.4	0.5	Mean	0	0.1	0.2	0.3	0.4	0.5	Mean	0	0.1	0.2	0.3	0.4	0.5	Mean
CSP	71.3	70.8	69.9	68.3	68.6	67.4	69.38	61.6	62.4	63.0	62.4	61.0	60.3	61.78	64.4	69.8	67.9	68.3	66.1	66.8	67.22
CSPmedian	69.8	71.0	69.1	70.4	69.3	69.5	69.85	62.8	62.6	63.3	63.2	61.0	62.4	62.55	61.4	65.1	69.4	65.2	67.3	67.4	65.97
TRCSP	71.1	71.1	69.0	71.1	68.3	69.6	70.03	62.1	62.5	63.6	64.1	64.5	61.7	63.08	65.6	68.8	70.3	68.2	68.6	68.2	68.28
CSPmcd	64.2	70.3	71.1	67.9	67.3	65.6	67.73	61.4	63.8	64.3	62.6	64.0	61.0	62.85	69.1	67.7	70.2	68.8	69.8	67.9	68.92
CSPweight	64.5	63.8	64.6	65.1	62.6	63.3	63.98	66.6	63.8	62.3	62.2	62.6	62.3	63.30	61.0	63.5	61.4	61.4	61.1	60.4	61.47
CSPmaxmin	69.0	71.5	71.5	69.1	68.9	67.5	69.58	62.5	63.8	64.6	63.4	63.6	61.6	63.25	67.5	69.6	70.1	69.4	69.0	68.3	68.98
CSP-L1	71.4	71.4	72.0	71.3	70.8	71.9	71.47	69.1	69.6	69.9	66.9	66.5	67.8	68.30	69.5	71.0	71.3	71.3	70.9	70.7	70.78
	Subject <i>A07E</i>							Subject <i>A08E</i>							Subject <i>A09E</i>						
Frequency	0	0.1	0.2	0.3	0.4	0.5	Mean	0	0.1	0.2	0.3	0.4	0.5	Mean	0	0.1	0.2	0.3	0.4	0.5	Mean
CSP	80.4	78.6	77.2	78.8	76.5	76.1	77.93	91.5	90.6	90.8	90.5	90.5	90.6	90.75	92.2	90.7	90.6	90.6	89.9	89.7	90.62
CSPmedian	75.3	77.5	79.1	77.9	77.5	75.2	77.08	92.0	91.1	91.1	90.7	91.0	91.1	91.17	92.7	91.1	90.5	89.8	90.0	89.8	90.65
TRCSP	80.9	79.4	78.4	78.4	78.3	76.1	78.58	92.7	92.1	91.6	91.8	91.3	91.3	91.80	92.7	91.4	91.2	91.1	90.4	90.3	91.18
CSPmcd	81.6	81.3	79.3	79.6	78.4	75.7	79.32	90.6	91.3	90.9	91.2	90.7	90.7	90.90	93.4	93.1	91.3	90.9	90.7	90.4	91.63
CSPweight	81.2	80.3	80.6	80.3	77.3	77.8	79.58	93.7	93.5	93.5	93.3	93.2	92.7	93.32	93.0	92.2	92.2	91.3	91.1	90.5	91.72
CSPmaxmin	82.1	80.8	79.3	80.0	78.4	76.9	79.58	92.1	92.0	92.0	91.7	91.9	91.6	91.88	93.8	92.9	91.6	91.9	91.5	90.9	92.10
CSP-L1	81.1	80.3	79.5	79.8	80.7	77.1	79.75	95.4	95.4	95.6	95.9	95.3	94.9	95.42	93.7	92.8	92.8	92.7	92.8	92.5	92.88

F. Discussion

We did not evaluate the performance of invariant CSP (iCSP) [3], which was also a robust CSP variant. The reason was that the noise covariance matrix, which needed to be computed by additional EEG signals, was not available. SCSP [1] aimed to select electrodes by sparse entries of filters, which was not intentionally designed for robust modeling and was excluded for comparison.

The performance of the standard CSP method deteriorated with the increasing occurrence frequencies of outliers on nearly all subjects except some subjects, e.g., *A04E*, *A05E*, and *A06E*. For subject *A02E*, all the methods produced slightly more than 50% recognition rates. In other words, they did not extract meaningful spatial filters regardless of outliers. The CSPmedian approach used the medians of variances, rather than the averages of variances as in CSP, to select filters. Nevertheless, it did not demonstrate good performance in resisting outliers. It may be due to that the median scores could not circumvent the effect of outliers when the number of outliers was large. Another reason may be that all the filters were already polluted by outliers before the selection using the criterion of median scores. The TRCSP had better performance than CSP, as agreed with [18]. This suggested that regularizing objective function of CSP was a feasible strategy to improve recognition results. The CSPmcd used part of samples, which were expected to be clean, to esti-

mate the covariance matrix. It had relatively better performance for some subjects. The percentage of selected samples, however, was difficult to be determined. Inappropriate percentage setting would deteriorate its performance. Even the fast algorithm of MCD was adopted [25], the computation of CSPmcd was rather time-consuming especially for large number of samples as in BCI domain. The CSPweight method could be actually viewed as a regularization technique, since it de-emphasized the contribution of outlier covariances by assigning them small weights. In some cases, it was useful. Its performance, however, was not stable. Overall, the weighting technique did not show encouraging advantage on the two datasets. The CSPmaxmin method demonstrated competitive recognition results. It had two parameters, i.e., radii of two tolerance balls, needed to be tuned appropriately. It eventually boiled down to the regularization of the covariance matrix [11].

We observe that, in most cases, the proposed CSP-L1 approach achieved the best performance on the EEG datasets without and with added outliers. CSP-L1 successfully resisted the affection of outliers. The nonparametric Friedman test [6] indicated statistical significance of the average recognition rates of the various CSP variants over subjects at the level 0.05. Posthoc multiple comparisons discovered that the average performance of CSP-L1 was significantly better than CSP and CSPmedian. It confirmed the superiority of CSP-L1 in learning spatial filters

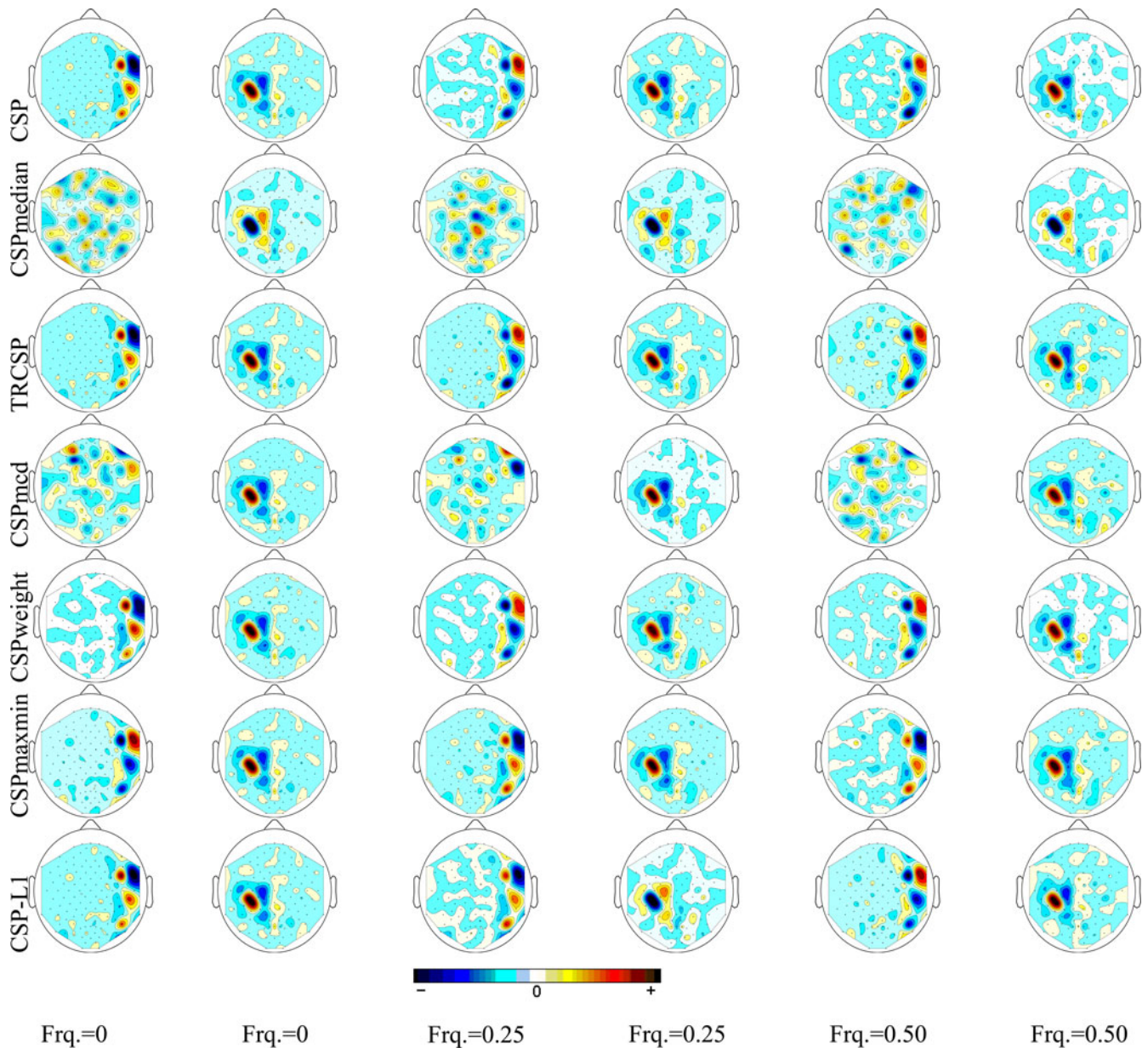


Fig. 4. Mappings of spatial filters learned by various CSP methods for subject *aa* on dataset IVa of BCI competition III in three cases: original data without outliers added ($\text{Frq.}=0$), and with outliers added at occurrence frequencies 0.25 and 0.50. In each case, two filters associated with the largest and the smallest eigenvalues were delineated.

from EEG data containing natural occurring or simulated outliers. The advantage of CSP-L1 is attributed to the utilization of L1-norm. By contrast, in the standard CSP method, variances were used as features for classification, since the CSP filters were optimized in terms of variances, i.e., the variance of one class was maximized while that of the other class was minimized. So, variance features, rather than L1-dispersion features, were discriminative with the CSP filters. Another interesting explanation was that CSP-L1 directly sought good spatial filters to constrain outliers rather than seeking good covariance matrices as a generic intermediate step, which agreed with Vapnik's statistical learning philosophy: "If you possess a restricted amount of information for solving some problem, try to solve the problem directly and never solve a more general problem

as an intermediate step. [26]," which was also demonstrated in [18]. Moreover, the objective function of CSP-L1 had some similarity with hinge loss.

Examples of mappings of spatial filters learned by various CSP methods for subject *aa* were shown in Fig. 4. In general, they looked somewhat messy, and put large values in some unexpected locations. The filters of CSP-L1 were relatively more relevant with the motor cortex areas. It implied that, like L2-norm, the one-order moment of L1-norm could be linked to the motor related rhythms. It was noted that the spatial filters of CSP-L1 remained relatively stable under varying occurrence frequencies of outliers. It was also interesting that the spatial filters of CSP-L1 were actually the spatial patterns, since they were orthonormal.

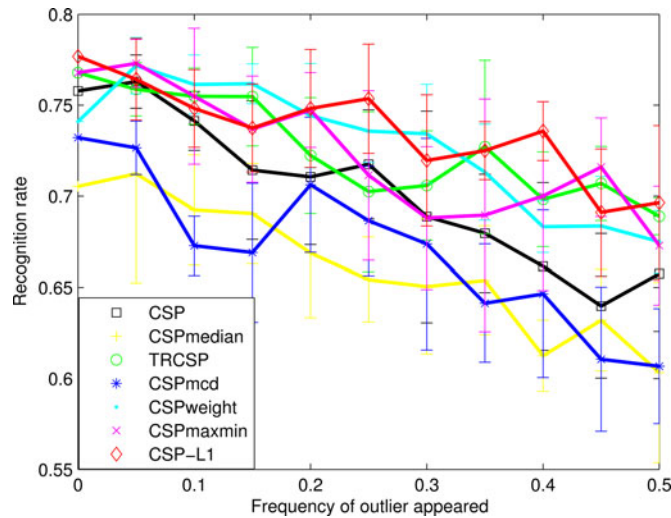


Fig. 5. Classification accuracies of subject *aa* on dataset IVa of BCI competition III using various CSP variants and the proposed CSP-L1 approach with increasing occurrence frequencies of outliers on both the training and the testing EEG data.

Our experiments were mainly to investigate the performance of spatial filters learned from noisy training data. To test the performance in the situation that both the training and the testing data were contaminated, we conducted a simple experiment on subject *aa*, as shown in Fig. 5. It was surprising that, compared with the results of Fig. 2 (a), the recognition results of Fig. 5 did not degrade. The reason may be that, the spatial filters were learned from contaminated data, and they did not favor clear testing data. On the contrast, the contaminated testing data achieved better recognition results in some case, since they were possibly more like the training data. If the training data were clear and the testing data were contaminated, then the seven methods yielded similar recognition results, since the spatial filters learned were similar.

The goal of this paper was to show the robustness of L1-norm applying to CSP: CSP-L1 produced better classification accuracy than other CSP variants under the same experimental settings. The classification performance may be improved by tuning the optimal number of spatial filters, combining multiple frequency bands, selecting the optimal time segment of each trial, and exploiting other advanced classifier. Only the outlier artifacts were considered in this paper. Other kind of noise and online testing are still needed to be investigated. We would study these issues in our future work.

V. CONCLUSION

In this paper, we proposed a new approach, called CSP-L1, for optimizing spatial filters. The approach was achieved by reformulating the conventional CSP expression using L1-norm rather than L2-norm. CSP-L1 was a robust extension to CSP due to the robust property of L1-norm. In other words, L1-norm could effectively relieve the impact of outliers. Computationally, we presented a simple but efficient iterative algorithm that monotonically increased the objective value of CSP-L1. The effectiveness of the proposed CSP-L1 method was shown by

the classification of two motor imagery tasks on two datasets of BCI competitions. We pointed out that, to find the globally maximization solution of CSP-L1, it was better to try different initializations. Also, the parameter η was needed to be carefully tuned. We are studying these issues theoretically and practically.

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